```
from google.colab import files
files.upload()
```

Choose files No file chosen Upload widget is only available when the cell has been executed in the current browser session. Please rerun this cell to enable.

```
Saving kaggle.json to kaggle.json
{'kaggle.json': b'{"username":"shamik777","key":"821f0db681fc008e84191731e32d5135"}'}
```

```
import os

os.makedirs('/root/.kaggle', exist_ok=True)
!mv kaggle.json /root/.kaggle/
```

```
!chmod 600 /root/.kaggle/kaggle.json
```

```
!pip install kaggle
```

```
Requirement already satisfied: kaggle in /usr/local/lib/python3.12/dist-packages (2.0.1)
Requirement already satisfied: bleach in /usr/local/lib/python3.12/dist-packages (from kaggle) (6.3.0)
Requirement already satisfied: kagglesdk<1.0,>=0.1.17 in /usr/local/lib/python3.12/dist-packages (from kaggle) (0
Requirement already satisfied: packaging in /usr/local/lib/python3.12/dist-packages (from kaggle) (26.0)
Requirement already satisfied: protobuf in /usr/local/lib/python3.12/dist-packages (from kaggle) (5.29.6)
Requirement already satisfied: python-dateutil in /usr/local/lib/python3.12/dist-packages (from kaggle) (2.9.0.po
Requirement already satisfied: python-slugify in /usr/local/lib/python3.12/dist-packages (from kaggle) (8.0.4)
Requirement already satisfied: requests in /usr/local/lib/python3.12/dist-packages (from kaggle) (2.32.4)
Requirement already satisfied: tqdm in /usr/local/lib/python3.12/dist-packages (from kaggle) (4.67.3)
Requirement already satisfied: urllib3>=1.15.1 in /usr/local/lib/python3.12/dist-packages (from kaggle) (2.5.0)
Requirement already satisfied: webencodings in /usr/local/lib/python3.12/dist-packages (from bleach->kaggle) (0.5
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from python-dateutil->kaggle)
Requirement already satisfied: text-unidecode>=1.3 in /usr/local/lib/python3.12/dist-packages (from python-slugif
Requirement already satisfied: charset_normalizer<4,>=2 in /usr/local/lib/python3.12/dist-packages (from requests
Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.12/dist-packages (from requests->kaggle) (3
Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/python3.12/dist-packages (from requests->kagg
```

```
!kaggle datasets list
```

```
ref                                                             title
--------------------------------------------------------------  --------------------------------------------------
algozee/teenager-menthal-healy                                  Social Media Impact on Teen Mental Health
nalisha/job-salary-prediction-dataset                           Job Salary Prediction Dataset
meruvakodandasuraj/youtube-trending-videos-20202026             YouTube Trending Videos 2020–2026
mubashirsidiki/student-academic-performance-500-students        Student Academic Performance - 500 Students
sumeakash/impact-of-social-media-on-health                      Impact of Social Media on Health
ihasan88/car-price-prediction-and-vehicle-specifications        Car Price Prediction and Vehicle Specifications
guriya79/how-ai-is-changing-student-life                        How AI is Changing Student Life
sohaibdevv/ai-and-student-life-2026-the-new-normal              How AI is Changing Life | AI and Student Life 2026
jajidhasan/mental-health                                        Mental Health
ssssws/global-inflation-dynamics-post-covid-20202024            Global Inflation Dynamics Post-COVID (2020–2024)
sumeakash/ai-impact-on-job-sector                               AI Impact on Job Sector
ssssws/chocolate-sales-dataset-2023-2024                        Chocolate Sales Dataset 2023 - 2024
nilesh2042/suzuki-stock-data                                    Suzuki Motor Corporation Stock Data (2015–2026)
jimarahman/smartphone-usage-and-addiction-analysis-dataset      Smartphone Usage and Addiction Analysis Dataset
kienngx/nemotron-30b-competition-trainingdata-cot-labels        Nemotron-30B competition trainingdata (CoT+labels)
algozee/smartphone-addiction-prediction-data                    Smartphone Addiction Prediction Data
yusufmurtaza01/youtube-top-channels-2026                        Top YouTube Channels: Global & Country-Wise (2026)
borovai0/student-performance-analytics-dataset                  Student Performance Analytics Dataset
belbino/us-tariff-and-trade-war-impact-dataset-2018-present     US Tariff & Trade War Impact Dataset 2018-Present
kanchana1990/cpsc-product-recall-intelligence                   CPSC Product Recall Intelligence
```

```
import kagglehub

# Download latest version
path = kagglehub.dataset_download("mateuszbuda/lgg-mri-segmentation")

print("Path to dataset files:", path)
```

```
Using Colab cache for faster access to the 'lgg-mri-segmentation' dataset.
Path to dataset files: /kaggle/input/lgg-mri-segmentation
```

```
! pip install torchview
```

```
Collecting torchview
  Downloading torchview-0.2.7-py3-none-any.whl.metadata (13 kB)
Requirement already satisfied: graphviz in /usr/local/lib/python3.12/dist-packages (from torchview) (0.21)
Downloading torchview-0.2.7-py3-none-any.whl (26 kB)
Installing collected packages: torchview
Successfully installed torchview-0.2.7
```

```
!pip install torchinfo
```

```
Collecting torchinfo
  Downloading torchinfo-1.8.0-py3-none-any.whl.metadata (21 kB)
Downloading torchinfo-1.8.0-py3-none-any.whl (23 kB)
Installing collected packages: torchinfo
Successfully installed torchinfo-1.8.0
```

```
from torchinfo import summary
```

```
from torchview import draw_graph
```

```
import random
import os
import glob
import time
import warnings

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

import torch
import torchvision
import torch.optim.lr_scheduler as lr_scheduler

from torch import nn
from torch.utils.data import (Dataset, DataLoader)
from torchvision import transforms
from torchinfo import summary
from torchview import draw_graph

import albumentations as A
from albumentations.pytorch import ToTensorV2

from PIL import Image
from tqdm.notebook import tqdm
from typing import Dict, List, Tuple

from sklearn.model_selection import train_test_split
from sklearn.metrics import (
    classification_report, precision_recall_fscore_support,
    accuracy_score, f1_score, matthews_corrcoef,
    confusion_matrix, ConfusionMatrixDisplay
)

plt.style.use("dark_background")
```

```
class CFG:
    DEVICE = 'cuda' if torch.cuda.is_available() else 'cpu'
    NUM_DEVICES = torch.cuda.device_count()
    NUM_WORKERS = os.cpu_count()

    NUM_CLASSES = 2
    EPOCHS = 100
    BATCH_SIZE = (
        32 if torch.cuda.device_count() < 2
        else (32 * torch.cuda.device_count())
    )
    LR = 1e-4
    PATIENS = 8

    APPLY_SHUFFLE = True
    SEED = 768
    HEIGHT = 224
    WIDTH = 224
    CHANNELS = 3
    IMAGE_SIZE = (224, 224, 3)

    # Define paths
    DATASET_PATH = "/kaggle/input/lgg-mri-segmentation/"
    TRAIN_PATH = '/kaggle/input/lgg-mri-segmentation/kaggle_3m/'

# Mute warnings
warnings.filterwarnings('ignore')
```

```
# Print current package versions
print('Package versions:')
print('*'*26)
```

```
print(f'torch \t\t - {torch.__version__}')
print(f'torchvision \t - {torchvision.__version__}')
```

```
Package versions:
**************************
torch            - 2.10.0+cu128
torchvision      - 0.25.0+cu128
```

```
%%time
dataset_images = glob.glob(f"{CFG.TRAIN_PATH}**/*.tif")
dataset_images[:5]
```

```
CPU times: user 20.6 ms, sys: 27.3 ms, total: 47.8 ms
Wall time: 987 ms
['/kaggle/input/lgg-mri-segmentation/kaggle_3m/TCGA_DU_7010_19860307/TCGA_DU_7010_19860307_45.tif',
 '/kaggle/input/lgg-mri-segmentation/kaggle_3m/TCGA_DU_7010_19860307/TCGA_DU_7010_19860307_56_mask.tif',
 '/kaggle/input/lgg-mri-segmentation/kaggle_3m/TCGA_DU_7010_19860307/TCGA_DU_7010_19860307_57.tif',
 '/kaggle/input/lgg-mri-segmentation/kaggle_3m/TCGA_DU_7010_19860307/TCGA_DU_7010_19860307_33.tif',
 '/kaggle/input/lgg-mri-segmentation/kaggle_3m/TCGA_DU_7010_19860307/TCGA_DU_7010_19860307_27.tif']
```

```
def get_sample_patient_id(image_paths):
    return [(_.split('/')[-2:][0]) for _ in image_paths]


def get_sample_number(image_paths):
    sample_numbers = []
    is_mask = []

    for path in image_paths:
        path_list = path.split('/')[-2:][1].split('_')

        if 'mask.tif' in path_list:
            sample_numbers.append(int(path_list[-2]))
            is_mask.append(1)
        else:
            sample_numbers.append(int(path_list[-1].replace('.tif', '')))
            is_mask.append(0)

    return sample_numbers, is_mask


def build_df(image_paths):
    sample_numbers, mask_label = get_sample_number(image_paths)
    # Create dataframe
    df = pd.DataFrame({
        'id'        : sample_numbers,
        'patient'   : get_sample_patient_id(image_paths),
        'image_path': image_paths,
        'is_mask'   : mask_label
    })

    # Return df
    return df
```

```
dataset_df = (
    build_df(dataset_images)
    .sort_values(by=['id', 'patient', 'image_path'])
    .reset_index(drop=True)
)

dataset_df
```

	id	patient	image_path	is_mask
0	1	TCGA_CS_4941_19960909	/kaggle/input/lgg-mri-segmentation/kaggle_3m/T...	0
1	1	TCGA_CS_4941_19960909	/kaggle/input/lgg-mri-segmentation/kaggle_3m/T...	1
2	1	TCGA_CS_4942_19970222	/kaggle/input/lgg-mri-segmentation/kaggle_3m/T...	0
3	1	TCGA_CS_4942_19970222	/kaggle/input/lgg-mri-segmentation/kaggle_3m/T...	1
4	1	TCGA_CS_4943_20000902	/kaggle/input/lgg-mri-segmentation/kaggle_3m/T...	0
...	...	...	...	...
7853	87	TCGA_HT_A61B_19991127	/kaggle/input/lgg-mri-segmentation/kaggle_3m/T...	1
7854	88	TCGA_HT_A61A_20000127	/kaggle/input/lgg-mri-segmentation/kaggle_3m/T...	0
7855	88	TCGA_HT_A61A_20000127	/kaggle/input/lgg-mri-segmentation/kaggle_3m/T...	1
7856	88	TCGA_HT_A61B_19991127	/kaggle/input/lgg-mri-segmentation/kaggle_3m/T...	0
7857	88	TCGA_HT_A61B_19991127	/kaggle/input/lgg-mri-segmentation/kaggle_3m/T...	1

7858 rows × 4 columns

```
# Group Samples by mask status
grouped_df = dataset_df.groupby(by='is_mask')

# Seperate Images from Masks
images_df, mask_df = (
    grouped_df.get_group(0).drop('is_mask', axis=1).reset_index(drop=True),
    grouped_df.get_group(1).drop('is_mask', axis=1).reset_index(drop=True)
)

# Rename path column in mask df
mask_df = mask_df.rename({'image_path': 'mask_path'}, axis=1)

# View mask df
mask_df
```

	id	patient	mask_path
0	1	TCGA_CS_4941_19960909	/kaggle/input/lgg-mri-segmentation/kaggle_3m/T...
1	1	TCGA_CS_4942_19970222	/kaggle/input/lgg-mri-segmentation/kaggle_3m/T...
2	1	TCGA_CS_4943_20000902	/kaggle/input/lgg-mri-segmentation/kaggle_3m/T...
3	1	TCGA_CS_4944_20010208	/kaggle/input/lgg-mri-segmentation/kaggle_3m/T...
4	1	TCGA_CS_5393_19990606	/kaggle/input/lgg-mri-segmentation/kaggle_3m/T...
...	...	...	...
3924	86	TCGA_HT_A61B_19991127	/kaggle/input/lgg-mri-segmentation/kaggle_3m/T...
3925	87	TCGA_HT_A61A_20000127	/kaggle/input/lgg-mri-segmentation/kaggle_3m/T...
3926	87	TCGA_HT_A61B_19991127	/kaggle/input/lgg-mri-segmentation/kaggle_3m/T...
3927	88	TCGA_HT_A61A_20000127	/kaggle/input/lgg-mri-segmentation/kaggle_3m/T...
3928	88	TCGA_HT_A61B_19991127	/kaggle/input/lgg-mri-segmentation/kaggle_3m/T...

3929 rows × 3 columns

```
def _load(image_path, as_tensor=True):
    image = Image.open(image_path)
    return np.array(image).astype(np.float32) / 255.


def generate_label(mask_path, load_fn):
    mask = load_fn(mask_path)
    if mask.max() > 0:
        return 1 # Brain Tumor Present
    return 0 # Normal
```

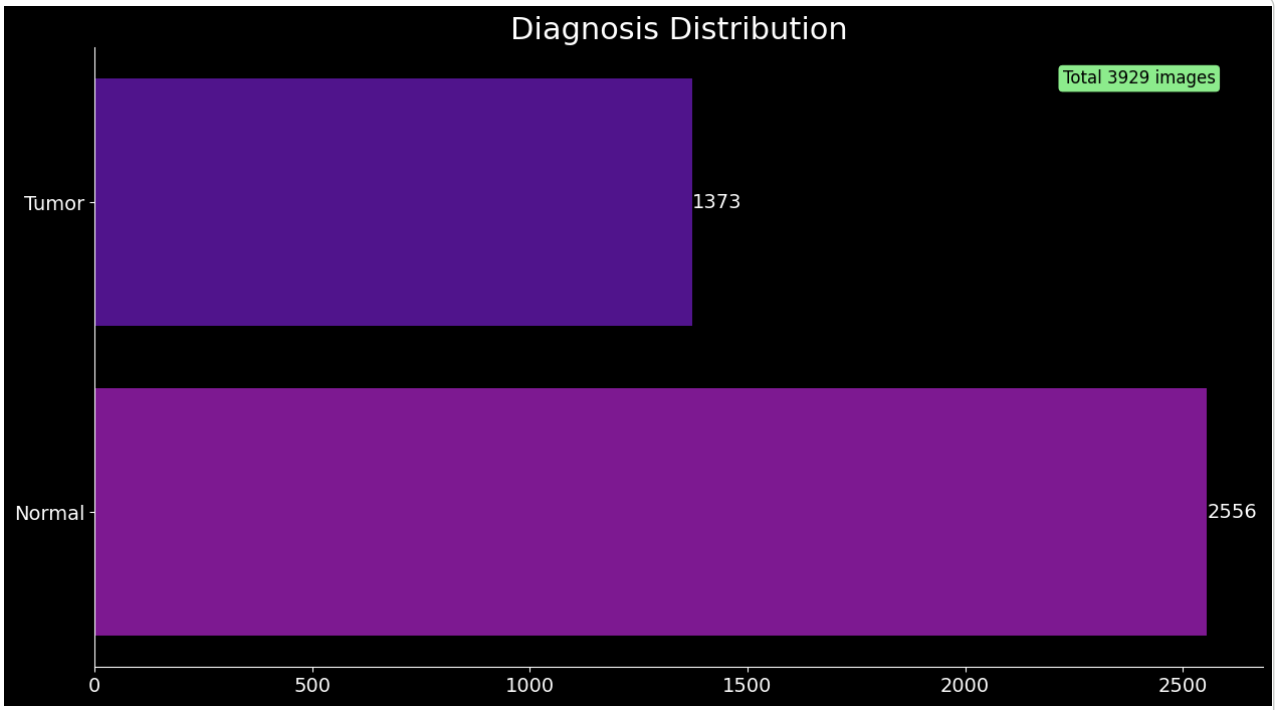
```
# Merge images and mask df's
ds = images_df.merge(
    mask_df,
    on=['id', 'patient'],
    how='left'
)

# Generate MRI Label
```

```
ds['diagnosis'] = [generate_label(_, _load) for _ in tqdm(ds['mask_path'])]
ds
```

100% 3929/3929 [00:23<00:00, 152.15it/s]

	id	patient	image_path	mask_path	diagnosis
0	1	TCGA_CS_4941_19960909	/kaggle/input/lgg-mri-segmentation/kaggle_3m/T...	/kaggle/input/lgg-mri-segmentation/kaggle_3m/T...	0
1	1	TCGA_CS_4942_19970222	/kaggle/input/lgg-mri-segmentation/kaggle_3m/T...	/kaggle/input/lgg-mri-segmentation/kaggle_3m/T...	0
2	1	TCGA_CS_4943_20000902	/kaggle/input/lgg-mri-segmentation/kaggle_3m/T...	/kaggle/input/lgg-mri-segmentation/kaggle_3m/T...	0
3	1	TCGA_CS_4944_20010208	/kaggle/input/lgg-mri-segmentation/kaggle_3m/T...	/kaggle/input/lgg-mri-segmentation/kaggle_3m/T...	0
4	1	TCGA_CS_5393_19990606	/kaggle/input/lgg-mri-segmentation/kaggle_3m/T...	/kaggle/input/lgg-mri-segmentation/kaggle_3m/T...	0
...	...	...	...	...	...
3924	86	TCGA_HT_A61B_19991127	/kaggle/input/lgg-mri-segmentation/kaggle_3m/T...	/kaggle/input/lgg-mri-segmentation/kaggle_3m/T...	0
3925	87	TCGA_HT_A61A_20000127	/kaggle/input/lgg-mri-segmentation/kaggle_3m/T...	/kaggle/input/lgg-mri-segmentation/kaggle_3m/T...	0

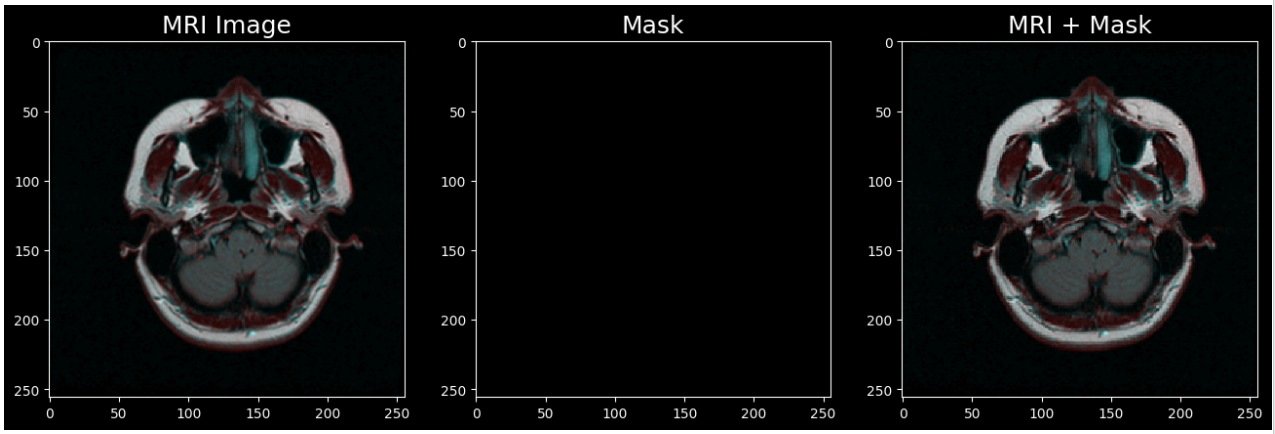
```
# View Diagnosis Distribution
fig = plt.figure(figsize=(15, 8))
plt.title('Diagnosis Distribution', fontsize=22);

diagnosis_distribution = ds['diagnosis'].value_counts().sort_values()
diagnosis_map = {0: 'Normal', 1: 'Tumor'}

bar_plot = sns.barplot(
    x=diagnosis_distribution.values,
    y=[diagnosis_map[_] for _ in diagnosis_distribution.keys()],
    palette=sns.color_palette("plasma"),
    orient="h"
);

for container in bar_plot.containers:
    bar_plot.bar_label(container, fmt='%.0f', fontsize=14);

plt.xticks(fontsize=14);
plt.yticks(fontsize=14);

text_params = {
    'x'       : 2400,
    'y'       : -0.4,
    's'       : f"Total {len(ds)} images",
    'size'    : 12,
    'color'   : "black",
    'ha'      : "center",
    'va'      : "center",
    'bbox'    : dict(boxstyle="round", fc=("lightgreen"),ec=("black"))
}
plt.text(**text_params);

sns.despine();
```



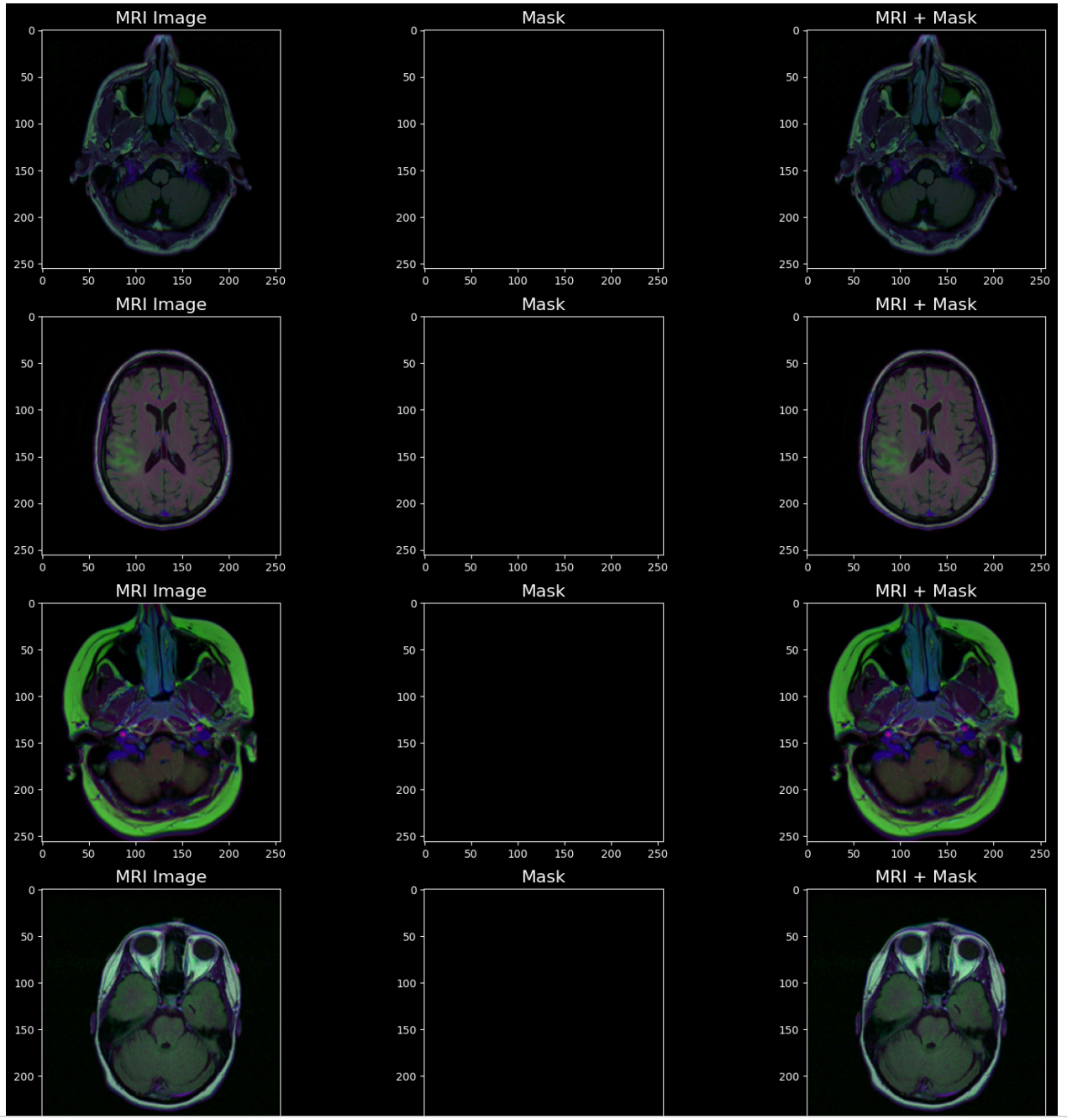

```
def view_sample(image, mask, color_map='rgb', fig_size=(16, 8), title_size=18):
    fig = plt.figure(figsize=fig_size);
    fig.tight_layout();
    fig.subplots_adjust(top=0.95);

    # Plot MRI Image
    plt.subplot(1, 3, 1);
    plt.title(f'MRI Image', fontsize=title_size)
    if color_map=='rgb':
        plt.imshow(image)
    else:
        plt.imshow(image, cmap=color_map)

    # Plot Mask
    plt.subplot(1, 3, 2);
    plt.title(f'Mask', fontsize=title_size)
    plt.imshow(mask, cmap='gray');

    # Plot MRI w. Mask
    plt.subplot(1, 3, 3);
    plt.title(f'MRI + Mask', fontsize=title_size)
    gen_mask = np.dstack([mask*0.1, mask*0.4, mask*0.1])

    if color_map=='rgb':
        plt.imshow(image + gen_mask, interpolation='none')
    else:
        plt.imshow(image + gen_mask, interpolation='none', cmap=color_map)

#     plt.imshow(mask, cmap='jet', interpolation='none', alpha=0.4);

    return
```

```
# Select random sample from train_df
idx = random.sample(ds.index.to_list(), 1)[0]

# Load the random sample and label
sample_image, sample_mask = _load(ds.image_path[idx]), _load(ds.mask_path[idx])

# View the random sample
view_sample(
    sample_image,
    sample_mask,
```

```
    color_map='rgb',
)
```




```
def view_mulitiple_samples(df, sample_loader, count=10, color_map='rgb', fig_size=(14, 10), title_size=16):
    idx = random.sample(df.index.to_list(), count)

    fig = plt.figure(figsize=fig_size);
    fig.tight_layout();
    fig.subplots_adjust(top=0.95);

    for row, _ in enumerate(idx):
        image, mask = (
            sample_loader(df.image_path[_]),#.permute(1, 2, 0),
            sample_loader(df.mask_path[_])#.permute(1, 2, 0)
        )

        idx = 3 * (row + 1)
        # Plot MRI Image
        plt.subplot(count+1, 3, idx + 1);
        plt.title(f'MRI Image', fontsize=title_size);
        if color_map=='rgb':
            plt.imshow(image)
        else:
            plt.imshow(image, cmap=color_map)

        # Plot Mask
        plt.subplot(count+1, 3, idx + 2);
        plt.title(f'Mask', fontsize=title_size)
        plt.imshow(mask, cmap='gray');

        # Plot MRI w. Mask
        plt.subplot(count+1, 3, idx + 3);
        gen_mask = np.dstack([mask*0.1, mask*0.4, mask*0.1])
        plt.title(f'MRI + Mask', fontsize=title_size)
        if color_map=='rgb':
            plt.imshow(image + gen_mask)
        else:
            plt.imshow(image + gen_mask, cmap=color_map)

#         plt.imshow(mask, cmap='gray', alpha=0.5);

    return

# View 6 random sample images
view_mulitiple_samples(
    ds, _load,
    count=6, color_map='mako',
    fig_size=(18, 30)
)
```




```
# Create Train split
train_split_idx, data_split_idx, _, _ = (
    train_test_split(
        ds.index,
        ds.diagnosis,
        test_size=0.30,
        stratify=ds.diagnosis,
        random_state=CFG.SEED
    )
)

# Get training and remaining data
train_df = ds.iloc[train_split_idx].reset_index(drop=True)
data_df = ds.iloc[data_split_idx].reset_index(drop=True)

# View shapes
train_df.shape, data_df.shape
```

```
((2250, 5), (1125, 5))
```

```
# Create Val/Test split
val_split_idx, test_split_idx, _, _ = (
    train_test_split(
        data_df.index,
        data_df.diagnosis,
        test_size=0.65,
        stratify=data_df.diagnosis,
        random_state=CFG.SEED
    )
```

```
)

# Get training and remaining data
val_df = data_df.iloc[val_split_idx].reset_index(drop=True)
test_df = data_df.iloc[test_split_idx].reset_index(drop=True)

# View shapes
val_df.shape, test_df.shape
```

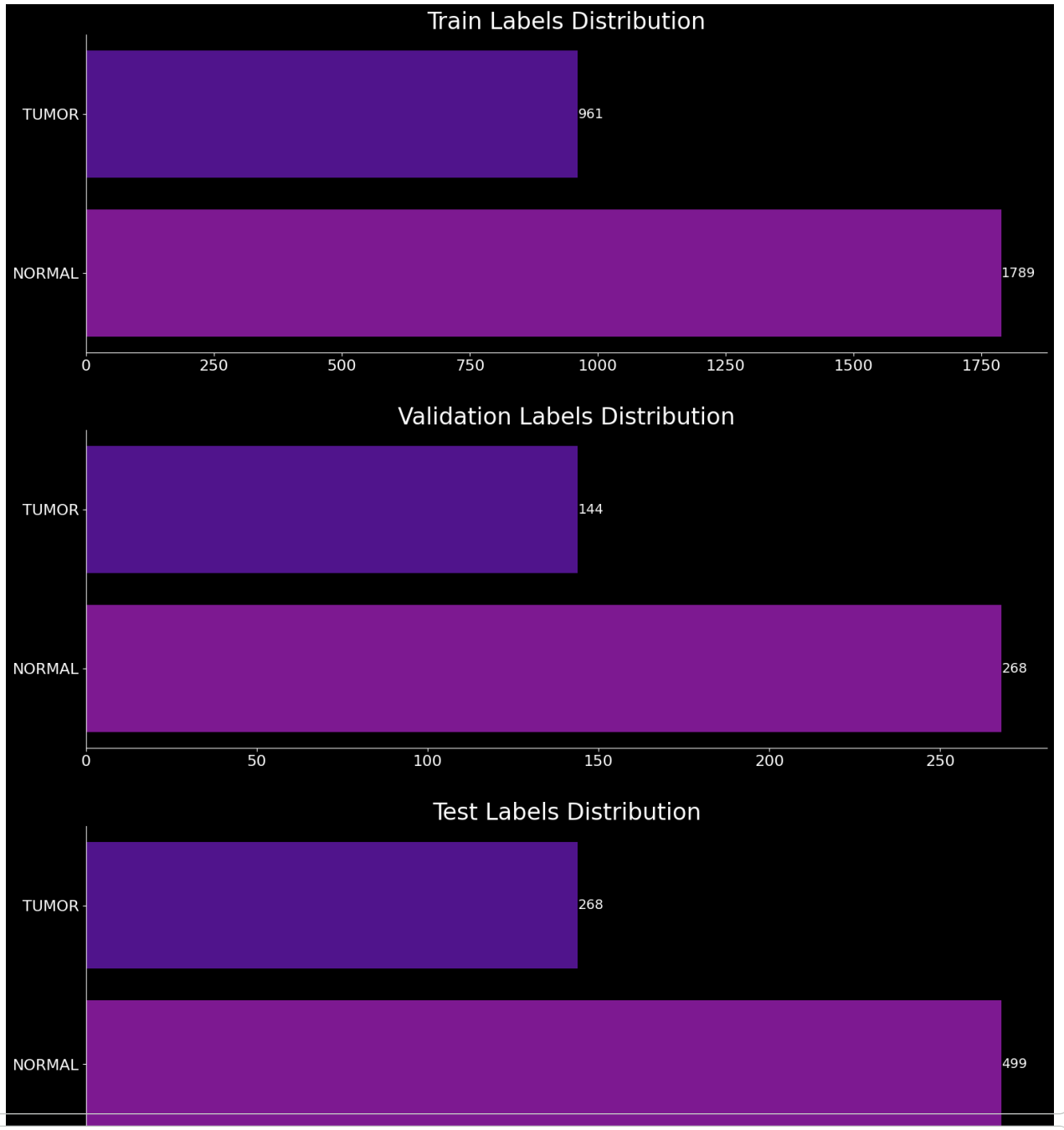
```
((412, 5), (767, 5))
```

```
train_size = len(train_df)
val_size = len(val_df)
test_size = len(test_df)
total = train_size + val_size + test_size

# View the counts
print(f'train samples count:\t\t{train_size}\t({(100 * train_size/total):.2f}%)')
print(f'validation samples count:\t{val_size}\t({(100 * val_size/total):.2f}%)')
print(f'test samples count:\t\t{test_size}\t({(100 * test_size/total):.2f}%)')
print('==================================================')
print(f'TOTAL:\t\t\t\t{total}\t({(100 * total/total):.2f}%)')
```

```
train samples count:            2750    (69.99%)
validation samples count:       412     (10.49%)
test samples count:             767     (19.52%)
==================================================
TOTAL:                          3929    (100.00%)
```

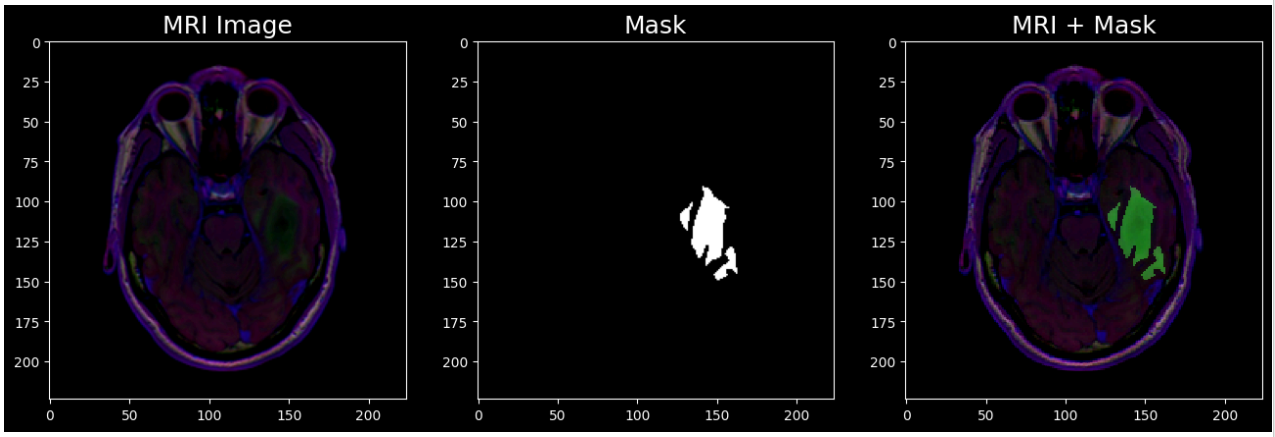
```
fig, (ax1, ax2, ax3) = plt.subplots(3, figsize=(16, 18))

# Set the spacing between subplots
fig.tight_layout(pad=6.0)
sns.despine();


# Plot Train Labels Distribution
ax1.set_title('Train Labels Distribution', fontsize=24)
train_distribution = train_df['diagnosis'].value_counts().sort_values()
labels = {0: 'NORMAL', 1: 'TUMOR'}
bar_plot = sns.barplot(
    x=train_distribution.values,
    y=[labels[_] for _ in train_distribution.keys()],
    orient="h", palette=sns.color_palette("plasma"),
    ax=ax1
);

for container in bar_plot.containers:
    bar_plot.bar_label(container, fmt='%.0f', fontsize=14);

# Update axes tick parameters
ax1.tick_params(
    axis='both', which='major',
    labelsize=16
);

# Plot Validation Labels Distribution
ax2.set_title('Validation Labels Distribution', fontsize=24)
val_distribution = val_df['diagnosis'].value_counts().sort_values()

bar_plot = sns.barplot(
    x=val_distribution.values,
    y=[labels[_] for _ in val_distribution.keys()],
    orient="h", palette=sns.color_palette("plasma"),
    ax=ax2
);

for container in bar_plot.containers:
    bar_plot.bar_label(container, fmt='%.0f', fontsize=14);

# Update axes tick parameters
ax2.tick_params(
    axis='both', which='major',
    labelsize=16
);

# Plot Test Labels Distribution
ax3.set_title('Test Labels Distribution', fontsize=24)
test_distribution = test_df['diagnosis'].value_counts().sort_values()

bar_plot = sns.barplot(
    x=test_distribution.values,
```

```
    y=[labels[_] for _ in test_distribution.keys()],
    orient="h", palette=sns.color_palette("plasma"),
    ax=ax3
);

for container in bar_plot.containers:
    bar_plot.bar_label(container, fmt='%.0f', fontsize=14);

# Update axes tick parameters
ax3.tick_params(
    axis='both', which='major',
    labelsize=16
);
```




```
# Augment train data
train_transforms = A.Compose([
    A.Resize(CFG.HEIGHT, CFG.WIDTH, p=1.0),
    A.RandomBrightnessContrast(p=0.2),
    A.HorizontalFlip(p=0.5),
    A.VerticalFlip(p=0.5),
    ToTensorV2(),
])

# Only reshape inference data
inference_transforms = A.Compose([
    A.Resize(CFG.HEIGHT, CFG.WIDTH, p=1.0),
    ToTensorV2(),
])
```

```
# Select random sample from train_df
idx = random.sample(ds.index.to_list(), 1)[0]

# Load the random sample and label
sample_image, sample_mask = _load(ds.image_path[idx]), _load(ds.mask_path[idx])

aug = train_transforms(image=sample_image, mask=sample_mask)

# View the random sample
view_sample(
    aug['image'].permute(1, 2, 0),
    aug['mask'].unsqueeze_(0).permute(1, 2, 0),
    color_map='mako',
)
```




```
class MRISegmetationDataset(Dataset):
    def __init__(self, df:pd.DataFrame, transform=None) -> None:
        super().__init__()

        self.mri_paths = df['image_path'].to_list()
        self.mask_paths = df['mask_path'].to_list()
        self.transform = transform

    def load_sample(self, index:str) -> Image.Image:
        image_path = self.mri_paths[index]
        mask_path = self.mask_paths[index]

        image, mask = Image.open(image_path), Image.open(mask_path)

        image = np.array(image).astype(np.float32) / 255.
        mask = np.array(mask).astype(np.float32) / 255.

        return image, mask

    def __len__(self) -> int:
        return self.mri_paths.__len__()

    def __getitem__(self, index:int) -> Tuple[torch.Tensor, torch.Tensor]:
        image, mask = self.load_sample(index)

        # Transform if necessary
        if self.transform:
            transformed = self.transform(image=image, mask=mask)
            return transformed['image'], transformed['mask'].unsqueeze_(0)
        else:
            transformed = ToTensorV2()(image=image, mask=mask)
            return transformed['image'], transformed['mask'].unsqueeze_(0)
```

```
# Build train dataset
train_ds = MRISegmetationDataset(
    train_df, transform=train_transforms
)

# Build validation dataset
val_ds = MRISegmetationDataset(
    val_df, transform=inference_transforms
)
```

```
# Build test dataset
test_ds = MRISegmetationDataset(
    test_df, transform=inference_transforms
)
```

```
# Build train dataloader
train_loader = DataLoader(
    dataset=train_ds,
    batch_size=CFG.BATCH_SIZE,
    num_workers=CFG.NUM_WORKERS,
    shuffle=CFG.APPLY_SHUFFLE
)

# Build validation dataloader
val_loader = DataLoader(
    dataset=val_ds,
    batch_size=CFG.BATCH_SIZE,
    num_workers=CFG.NUM_WORKERS,
    shuffle=False
)

# Build test dataloader
test_loader = DataLoader(
    dataset=test_ds,
    batch_size=CFG.BATCH_SIZE,
    num_workers=CFG.NUM_WORKERS,
    shuffle=False
)
```

```
def execute_epoch(
    model:torch.nn.Module,
    dataloader:torch.utils.data.DataLoader,
    optimizer:torch.optim.Optimizer,
    loss_fn:torch.nn.Module,
    device:torch.device) -> Tuple[float, float]:

    # Set model into training mode
    model.train()

    # Initialize train loss & accuracy
    train_loss, train_dice = 0, 0

    # Execute training loop over train dataloader
    for batch, (X, y) in enumerate(dataloader):
        # Load data onto target device
        X, y = X.to(device), y.to(device)

        # Feed-forward and compute metrics
        y_pred = model(X)
        loss = loss_fn(y_pred, y)
        train_loss += loss.item()

        # Reset Gradients & Backpropagate Loss
        optimizer.zero_grad()
        loss.backward()

        # Update Model Gradients
        optimizer.step()

        # Compute Batch Metrics
        predicted_class = torch.sigmoid(y_pred)
        predicted_class = (predicted_class > 0.5).float()

        eps = 1e-8
        train_dice += (
            (2 * (y * predicted_class).sum() + eps) /
            ((y + predicted_class).sum() + eps)
        ).cpu().item()


    # Compute Step Metrics
    train_loss = train_loss / len(dataloader)
    train_dice = train_dice / len(dataloader)

    return train_loss, train_dice
```

```
def evaluate(
    model:torch.nn.Module,
    dataloader:torch.utils.data.DataLoader,
    loss_fn:torch.nn.Module,
```

```
    device:torch.device) -> Tuple[float, float]:

    # Set model into eval mode
    model.eval()

    # Initialize eval loss & accuracy
    eval_loss, eval_dice = 0, 0

    # Active inferene context manager
    with torch.inference_mode():
        # Execute eval loop over dataloader
        for batch, (X, y) in enumerate(dataloader):
            # Load data onto target device
            X, y = X.to(device), y.to(device)

            # Feed-forward and compute metrics
            y_pred = model(X)
            loss = loss_fn(y_pred, y)
            eval_loss += loss.item()

            # Compute Batch Metrics
            predicted_class = torch.sigmoid(y_pred)
            predicted_class = (predicted_class > 0.5).float()

            eps = 1e-8
            eval_dice += (
                (2 * (y * predicted_class).sum() + eps) /
                ((y + predicted_class).sum() + eps)
            ).cpu().item()

    # Compute Step Metrics
    eval_loss = eval_loss / len(dataloader)
    eval_dice = eval_dice / len(dataloader)

    return eval_loss, eval_dice
```

```
def train(
    model:torch.nn.Module,
    train_dataloader:torch.utils.data.DataLoader,
    eval_dataloader:torch.utils.data.DataLoader,
    optimizer:torch.optim.Optimizer,
    scheduler:torch.optim.lr_scheduler,
    loss_fn:torch.nn.Module,
    epochs:int,
    device:torch.device) -> Dict[str, List]:

    # Initialize training session
    session = {
        'loss'            : [],
        'dice_score'      : [],
        'eval_loss'       : [],
        'eval_dice_score' : []
    }

    # Training loop
    for epoch in tqdm(range(epochs)):
        # Execute Epoch
        print(f'\nEpoch {epoch + 1}/{epochs}')
        train_loss, train_dice = execute_epoch(
            model,
            train_dataloader,
            optimizer,
            loss_fn,
            device
        )

        # Evaluate Model
        eval_loss, eval_dice = evaluate(
            model,
            eval_dataloader,
            loss_fn,
            device
        )

        # Execute schedular step
        current_lr = 0
        if scheduler:
            scheduler.step(eval_loss)
            current_lr = optimizer.param_groups[0]['lr']

        # Log Epoch Metrics
        log_text = f'loss: {train_loss:.4f} - dice_score: {train_dice:.4f} - eval_loss: {eval_loss:.4f} - eval_d
```

```
        if scheduler:
            print(log_text + f' - lr: {current_lr}')
        else:
            print(log_text)

        # Record Epoch Metrics
        session['loss'].append(train_loss)
        session['dice_score'].append(train_dice)
        session['eval_loss'].append(eval_loss)
        session['eval_dice_score'].append(eval_dice)

    # Return Session Metrics
    return session
```

```
def predict(
    model:nn.Module,
    sample_loader:torch.utils.data.DataLoader,
    device:torch.device,
    threshold:float=0.5) -> np.ndarray:

    # Set model into eval mode
    model.eval()

    predictions = []

    # Active inferene context manager
    with torch.inference_mode():
        # Execute eval loop over dataloader
        for batch, (X, y) in enumerate(tqdm(sample_loader)):
            # Load data onto target device
            X, y = X.to(device), y.to(device)

            # Feed-forward and compute metrics
            y_pred = model(X)

            # Compute Batch Metrics
            predicted_class = torch.sigmoid(y_pred)
            predicted_class = (predicted_class >= threshold).float()

            # Record prediction
            predictions.append(predicted_class.cpu().numpy())

    return np.vstack(predictions)
```

```
class DoubleConvBlock(nn.Module):
    def __init__(self, in_channels, out_channels):
        super(DoubleConvBlock, self).__init__()

        self.conv1 = nn.Conv2d(
            in_channels, out_channels,
            kernel_size=3, stride=1,
            padding=1, bias=False
        )
        self.conv2 = nn.Conv2d(
            out_channels, out_channels,
            kernel_size=3, stride=1,
            padding=1, bias=False
        )

        self.batchnorm1 = nn.BatchNorm2d(out_channels)
        self.batchnorm2 = nn.BatchNorm2d(out_channels)

        self.relu1 = nn.ReLU(inplace=True)
        self.relu2 = nn.ReLU(inplace=True)


    def forward(self, x):
        # First Convolution
        x = self.conv1(x)
        x = self.batchnorm1(x)
        x = self.relu1(x)

        # Second Convolution
        x = self.conv2(x)
        x = self.batchnorm2(x)

        return self.relu2(x)
```

```
class UNetModel(nn.Module):
    def __init__(
        self, in_channels:int=3,
        out_channels:int=1,
        block_sizes:Tuple[int]=(64, 128, 256, 512)
    ):
        super(UNetModel, self).__init__()

        # Initialise model encoder & decoder using torch ModuleLists
        self.encoder, self.decoder = nn.ModuleList(), nn.ModuleList()
        self.max_pool = nn.MaxPool2d(kernel_size=2, stride=2)

        # Create Encoder
        for block_size in block_sizes:
            self.encoder.append(DoubleConvBlock(in_channels, block_size))
            in_channels = block_size

        # Create Decoder
        for block_size in block_sizes[::-1]:
            self.decoder.append(
                nn.ConvTranspose2d(2 * block_size, block_size, kernel_size=2, stride=2)
            )
            self.decoder.append(DoubleConvBlock(2 * block_size, block_size))

        # Create Bottleneck
        last_blocksize = block_sizes[-1]
        self.bottleneck = DoubleConvBlock(last_blocksize, 2 * last_blocksize)

        # Create Output Layer
        self.output_conv = nn.Conv2d(block_sizes[0], out_channels, kernel_size=1)

    def forward(self, x):
        concatenations = []
        concatenations.append(x)

        # Propagate input downstream (Encode Input)
        for encoder_layer in self.encoder:
            x = encoder_layer(x)
            concatenations.append(x)
            x = self.max_pool(x)

        # Execute bottleneck
        x = self.bottleneck(x)
        concatenations = concatenations[::-1]

        # Propagate input upstream (Decode Input) & concatenate layers
        for _ in range(0, len(self.decoder), 2):
            x = self.decoder[_](x)
            encoder_layer = concatenations[_ // 2]

            # Concatenate corrensponding encoder layer to decoder layer output
            concat_layer = torch.cat(
                (encoder_layer, x), dim=1
            )

            x = self.decoder[_ + 1](concat_layer)

        # Return predicted logits
        return self.output_conv(x)
```

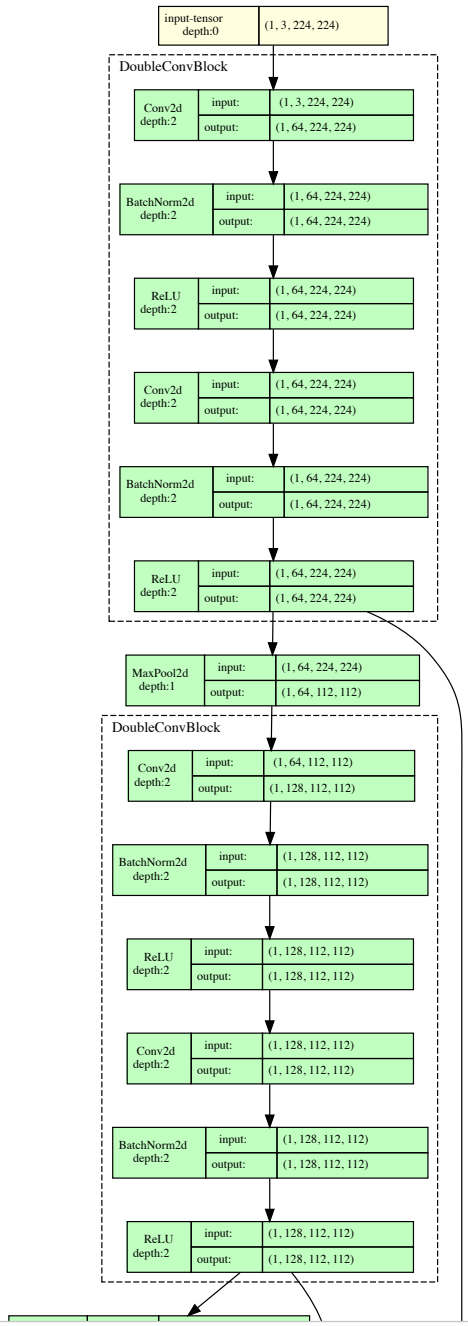
```
# Define U-Net Params
unet_params = {
    'in_channels'    : 3,
    'out_channels'   : 1,
    'block_sizes'    : (64, 128, 256, 512)
}

# Generate Model & Push to Device
unet_model = UNetModel(**unet_params).to(CFG.DEVICE)

# If using GPU T4 x2 setup, use this:
if CFG.NUM_DEVICES > 1:
    unet_model = nn.DataParallel(unet_model)
```

```
# View model summary
summary(
    model=unet_model,
    input_size=(CFG.BATCH_SIZE, CFG.CHANNELS, CFG.WIDTH, CFG.HEIGHT),
    col_names=["output_size", "num_params", "trainable"],
    col_width=20,
```

```
    row_settings=["var_names"]
)
```

```
==========================================================================================
Layer (type (var_name))                  Output Shape         Param #              Trainable
==========================================================================================
UNetModel (UNetModel)                    [32, 1, 224, 224]    --                   True
├─ModuleList (encoder)                   --                   (recursive)          True
│    └─DoubleConvBlock (0)               [32, 64, 224, 224]   --                   True
│    │    └─Conv2d (conv1)               [32, 64, 224, 224]   1,728                True
│    │    └─BatchNorm2d (batchnorm1)     [32, 64, 224, 224]   128                  True
│    │    └─ReLU (relu1)                 [32, 64, 224, 224]   --                   --
│    │    └─Conv2d (conv2)               [32, 64, 224, 224]   36,864               True
│    │    └─BatchNorm2d (batchnorm2)     [32, 64, 224, 224]   128                  True
│    │    └─ReLU (relu2)                 [32, 64, 224, 224]   --                   --
├─MaxPool2d (max_pool)                   [32, 64, 112, 112]   --                   --
├─ModuleList (encoder)                   --                   (recursive)          True
│    └─DoubleConvBlock (1)               [32, 128, 112, 112]  --                   True
│    │    └─Conv2d (conv1)               [32, 128, 112, 112]  73,728               True
│    │    └─BatchNorm2d (batchnorm1)     [32, 128, 112, 112]  256                  True
│    │    └─ReLU (relu1)                 [32, 128, 112, 112]  --                   --
│    │    └─Conv2d (conv2)               [32, 128, 112, 112]  147,456              True
│    │    └─BatchNorm2d (batchnorm2)     [32, 128, 112, 112]  256                  True
│    │    └─ReLU (relu2)                 [32, 128, 112, 112]  --                   --
├─MaxPool2d (max_pool)                   [32, 128, 56, 56]    --                   --
├─ModuleList (encoder)                   --                   (recursive)          True
│    └─DoubleConvBlock (2)               [32, 256, 56, 56]    --                   True
│    │    └─Conv2d (conv1)               [32, 256, 56, 56]    294,912              True
│    │    └─BatchNorm2d (batchnorm1)     [32, 256, 56, 56]    512                  True
│    │    └─ReLU (relu1)                 [32, 256, 56, 56]    --                   --
│    │    └─Conv2d (conv2)               [32, 256, 56, 56]    589,824              True
│    │    └─BatchNorm2d (batchnorm2)     [32, 256, 56, 56]    512                  True
│    │    └─ReLU (relu2)                 [32, 256, 56, 56]    --                   --
├─MaxPool2d (max_pool)                   [32, 256, 28, 28]    --                   --
├─ModuleList (encoder)                   --                   (recursive)          True
│    └─DoubleConvBlock (3)               [32, 512, 28, 28]    --                   True
│    │    └─Conv2d (conv1)               [32, 512, 28, 28]    1,179,648            True
│    │    └─BatchNorm2d (batchnorm1)     [32, 512, 28, 28]    1,024                True
│    │    └─ReLU (relu1)                 [32, 512, 28, 28]    --                   --
│    │    └─Conv2d (conv2)               [32, 512, 28, 28]    2,359,296            True
│    │    └─BatchNorm2d (batchnorm2)     [32, 512, 28, 28]    1,024                True
│    │    └─ReLU (relu2)                 [32, 512, 28, 28]    --                   --
├─MaxPool2d (max_pool)                   [32, 512, 14, 14]    --                   --
├─DoubleConvBlock (bottleneck)           [32, 1024, 14, 14]   --                   True
│    └─Conv2d (conv1)                    [32, 1024, 14, 14]   4,718,592            True
│    └─BatchNorm2d (batchnorm1)          [32, 1024, 14, 14]   2,048                True
│    └─ReLU (relu1)                      [32, 1024, 14, 14]   --                   --
│    └─Conv2d (conv2)                    [32, 1024, 14, 14]   9,437,184            True
│    └─BatchNorm2d (batchnorm2)          [32, 1024, 14, 14]   2,048                True
│    └─ReLU (relu2)                      [32, 1024, 14, 14]   --                   --
├─ModuleList (decoder)                   --                   --                   True
│    └─ConvTranspose2d (0)               [32, 512, 28, 28]    2,097,664            True
│    └─DoubleConvBlock (1)               [32, 512, 28, 28]    --                   True
│    │    └─Conv2d (conv1)               [32, 512, 28, 28]    4,718,592            True
│    │    └─BatchNorm2d (batchnorm1)     [32, 512, 28, 28]    1,024                True
│    │    └─ReLU (relu1)                 [32, 512, 28, 28]    --                   --
│    │    └─Conv2d (conv2)               [32, 512, 28, 28]    2,359,296            True
│    │    └─BatchNorm2d (batchnorm2)     [32, 512, 28, 28]    1,024                True
│    │    └─ReLU (relu2)                 [32, 512, 28, 28]    --                   --
│    └─ConvTranspose2d (2)               [32, 256, 56, 56]    524,544              True
│    └─DoubleConvBlock (3)               [32, 256, 56, 56]    --                   True
```

```
# Generate Model Computation Graph
model_graph = draw_graph(
    UNetModel(),
    input_size=(1, CFG.CHANNELS, CFG.WIDTH, CFG.HEIGHT),
    expand_nested=True
)

# View Model Architecture
model_graph.visual_graph
```




```
# Define Loss Function
loss_fn = nn.BCEWithLogitsLoss()

# Define Optimizer
optimizer = torch.optim.AdamW(
    unet_model.parameters(),
    lr=CFG.LR
)

# Define Scheduler
scheduler = lr_scheduler.ReduceLROnPlateau(
    optimizer=optimizer,
    mode='min',
    patience=CFG.PATIENS
)
```

```
print('Training U-Net Model')
print(f'Train on {len(train_df)} samples, validate on {len(val_df)} samples.')
print('----------------------------------')

# Generate training session config
session_config = {
    'model'               : unet_model,
    'train_dataloader'    : train_loader,
    'eval_dataloader'     : val_loader,
    'optimizer'           : optimizer,
    'scheduler'           : scheduler,
    'loss_fn'             : loss_fn,
```

```
    'epochs'              : CFG.EPOCHS,
    'device'              : CFG.DEVICE
}

# Execute Training Session
unet_session_history = train(**session_config)
```

```
Training U-Net Model
Train on 2750 samples, validate on 412 samples.
----------------------------------
100%                                                  100/100 [3:03:39<00:00, 110.21s/it]

Epoch 1/100
loss: 0.3529 - dice_score: 0.2951 - eval_loss: 0.2586 - eval_dice_score: 0.5372 - lr: 0.0001

Epoch 2/100
loss: 0.2281 - dice_score: 0.4627 - eval_loss: 0.2090 - eval_dice_score: 0.5517 - lr: 0.0001

Epoch 3/100
loss: 0.1901 - dice_score: 0.5220 - eval_loss: 0.1855 - eval_dice_score: 0.5703 - lr: 0.0001

Epoch 4/100
loss: 0.1628 - dice_score: 0.5596 - eval_loss: 0.1513 - eval_dice_score: 0.6441 - lr: 0.0001

Epoch 5/100
loss: 0.1410 - dice_score: 0.6090 - eval_loss: 0.1343 - eval_dice_score: 0.6297 - lr: 0.0001

Epoch 6/100
loss: 0.1214 - dice_score: 0.6428 - eval_loss: 0.1118 - eval_dice_score: 0.7145 - lr: 0.0001

Epoch 7/100
loss: 0.1052 - dice_score: 0.6761 - eval_loss: 0.0976 - eval_dice_score: 0.7330 - lr: 0.0001

Epoch 8/100
loss: 0.0921 - dice_score: 0.7011 - eval_loss: 0.0861 - eval_dice_score: 0.7190 - lr: 0.0001

Epoch 9/100
loss: 0.0820 - dice_score: 0.7077 - eval_loss: 0.0783 - eval_dice_score: 0.7261 - lr: 0.0001

Epoch 10/100
loss: 0.0726 - dice_score: 0.7324 - eval_loss: 0.0697 - eval_dice_score: 0.7311 - lr: 0.0001

Epoch 11/100
loss: 0.0653 - dice_score: 0.7362 - eval_loss: 0.0610 - eval_dice_score: 0.7808 - lr: 0.0001

Epoch 12/100
loss: 0.0589 - dice_score: 0.7461 - eval_loss: 0.0554 - eval_dice_score: 0.7779 - lr: 0.0001

Epoch 13/100
loss: 0.0538 - dice_score: 0.7405 - eval_loss: 0.0549 - eval_dice_score: 0.7403 - lr: 0.0001

Epoch 14/100
loss: 0.0480 - dice_score: 0.7760 - eval_loss: 0.0430 - eval_dice_score: 0.7773 - lr: 0.0001

Epoch 15/100
loss: 0.0443 - dice_score: 0.7735 - eval_loss: 0.0435 - eval_dice_score: 0.7887 - lr: 0.0001

Epoch 16/100
loss: 0.0406 - dice_score: 0.7737 - eval_loss: 0.0381 - eval_dice_score: 0.7956 - lr: 0.0001

Epoch 17/100
loss: 0.0375 - dice_score: 0.7840 - eval_loss: 0.0361 - eval_dice_score: 0.8027 - lr: 0.0001

Epoch 18/100
loss: 0.0314 - dice_score: 0.8172 - eval_loss: 0.0287 - eval_dice_score: 0.8338 - lr: 0.0001
```

```
from google.colab import drive
drive.mount('/content/drive')
```

```
Mounted at /content/drive
```

```
# Create Model directory


model_name = 'unet_model'
model_path = '/content/drive/MyDrive/model/'
os.makedirs(model_path, exist_ok=True)
```

```
# Save Model
torch.save(unet_model, model_path + model_name + '.pth')
```

```
Epoch 24/100
```

```
# Generate Test Masks
unet_test_masks = predict(
    unet_model, test_loader,
    CFG.DEVICE, threshold=0.5
)
```

```
loss: 0.0200 - dice_score: 0.8407 - eval_loss: 0.0185 - eval_dice_score: 0.8506 - lr: 0.0001
```

```
Epoch
loss: 0.0180 - dice_score: 0.8435 - eval_loss: 0.0187 - eval_dice_score: 0.8613 - lr: 0.0001
```

Performance Analysis

```
Epoch 28/100
loss: 0.0180 - dice_score: 0.8477 - eval_loss: 0.0170 - eval_dice_score: 0.8733 - lr: 0.0001
```

```
def plot_training_curves(history, fig_size=(20, 10)):

    loss = np.array(history['loss'])
    val_loss = np.array(history['eval_loss'])

    dice_coeff = np.array(history['dice_score'])
    val_dice_coeff = np.array(history['eval_dice_score'])

    epochs = range(len(history['loss']))

    fig, (ax1, ax2) = plt.subplots(2, 1, figsize=fig_size)

    # Plot loss
    ax1.plot(epochs, loss, label='training_loss', marker='o', color='C5');
    ax1.plot(epochs, val_loss, label='eval_loss', marker='o', color='C6');

    # Fill area between losses
    ax1.fill_between(epochs, loss, val_loss, where=(loss > val_loss), color='C5', alpha=0.4, interpolate=True);
    ax1.fill_between(epochs, loss, val_loss, where=(loss < val_loss), color='C6', alpha=0.4, interpolate=True);

    # Add Text & Formats
    ax1.set_title('Loss (Lower Means Better)', fontsize=22);
    ax1.set_xlabel('Epochs', fontsize=18);
    ax1.set_ylabel('Loss', fontsize=18);
    ax1.tick_params(axis='both', which='major', labelsize=14);
    ax1.legend(fontsize=14);

    # Plot metric
    ax2.plot(epochs, dice_coeff, label='training_dice_score', marker='o', color='C5');
    ax2.plot(epochs, val_dice_coeff, label='eval_dice_score', marker='o', color='C6');

    # Fill area between metrics
    ax2.fill_between(epochs, dice_coeff, val_dice_coeff, where=(dice_coeff > val_dice_coeff), color='C5', alpha=
    ax2.fill_between(epochs, dice_coeff, val_dice_coeff, where=(dice_coeff < val_dice_coeff), color='C6', alpha=

    # Add Text & Formats
    ax2.set_title('Dice Score (Higher Means Better)', fontsize=22);
    ax2.set_xlabel('Epochs', fontsize=18);
    ax2.set_ylabel('Dice Score', fontsize=18);
    ax2.tick_params(axis='both', which='major', labelsize=14);
    ax2.legend(fontsize=14);

    sns.despine();

    return
```

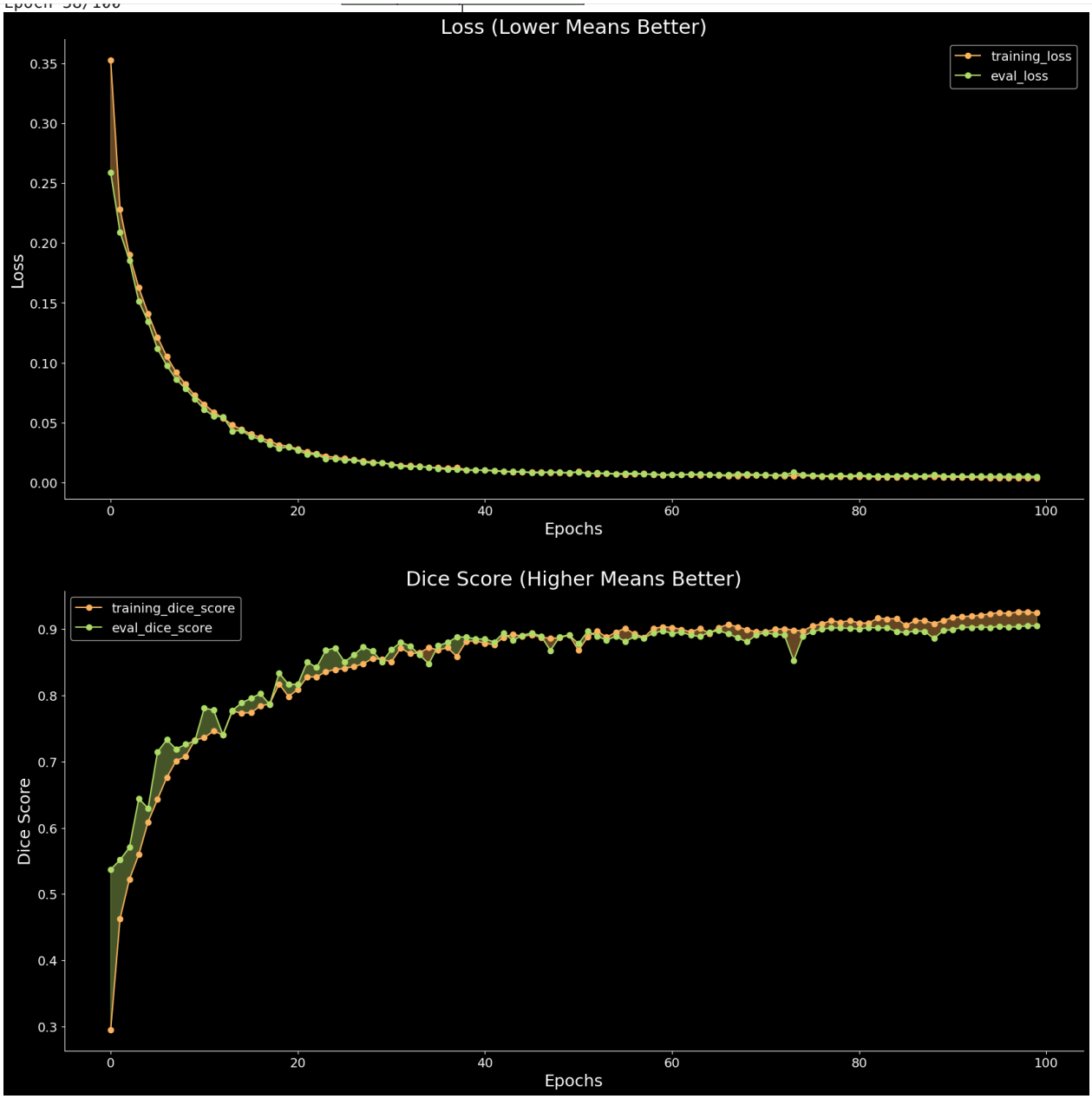
```
# Convert U-Net history dict to DataFrame
unet_session_history_df = pd.DataFrame(unet_session_history)
unet_session_history_df
```

	loss	dice_score	eval_loss	eval_dice_score
0	0.352871	0.295115	0.258617	0.537151
1	0.228095	0.462687	0.208959	0.551698
2	0.190073	0.522046	0.185493	0.570321
3	0.162762	0.559615	0.151298	0.644066
4	0.141009	0.608957	0.134253	0.629735
95	0.003897	0.924986	0.005210	0.904073
96	0.003988	0.923461	0.005195	0.903168
97	0.003869	0.926348	0.005188	0.903600
98	0.003856	0.926125	0.005188	0.905049
99	0.004040	0.924451	0.005173	0.904925

100 rows × 4 columns

```
# Plot U-Net Session Training History
plot_training_curves(
```

```
    unet_session_history,
    fig_size=(20, 20)
)
```




```
loss: 0.0052 - dice_score: 0.9049 - eval_loss: 0.0062 - eval_dice_score: 0.8960 - lr: 0.0001

Epoch 77/100
loss: 0.0051 - dice_score: 0.9079 - eval_loss: 0.0057 - eval_dice_score: 0.8999 - lr: 0.0001
```

```
# def accuracy(y_true, y_pred):
#     intersection = (y_true * y_pred).sum()
#     union = (y_true + y_pred).sum() - intersection
#     xor = (y_true == y_pred).sum()

#     return (xor / (union + xor - intersection))


def precision_(y_true, y_pred):
    intersection = (y_true * y_pred).sum()
    total_predicted_pixels = y_pred.sum()
    return (intersection / total_predicted_pixels).mean()


def recall_(y_true, y_pred):
    intersection = (y_true * y_pred).sum()
    total_true_pixels = y_true.sum()
    return (intersection / total_true_pixels).mean()


def dice_score(y_true, y_pred):
    eps = 1e-8
    intersection = (y_true * y_pred).sum()
    summation = (y_true + y_pred).sum()
```

```
    return ((2 * intersection) / (summation + eps))


def jaccard_index(y_true, y_pred):
    eps = 1e-8
    intersection = (y_true * y_pred).sum()
    union = (y_true + y_pred).sum() - intersection

    return (intersection / (union + eps))
```

```
def compute_metrics(
    model:nn.Module,
    sample_loader:torch.utils.data.DataLoader,
    device:torch.device) -> np.ndarray:

    # Initiate Metrics Dict
    metrics = {
        'accuracy'      : [],
        'precision'     : [],
        'recall'        : [],
        'dice_score'    : [],
        'IoU'           : []
    }

    # Set model into eval mode
    model.eval()

    # Active inferene context manager
    with torch.inference_mode():
        # Execute eval loop over dataloader
        for batch, (X, y) in enumerate(tqdm(sample_loader)):
            # Load data onto target device
            X, y = X.to(device), y.to(device)

            # Feed-forward Input
            y_pred = model(X)

            # Generate Predicted Masks
            predicted_class = torch.sigmoid(y_pred)
            predicted_class = (predicted_class > 0.3).float()

            # Compute Batch Metrics For Each Mask
            for true_mask, pred_mask in zip(y, predicted_class):
                acc = ((true_mask == pred_mask).sum() / torch.numel(true_mask)).cpu().item()
                prec = precision_(true_mask, pred_mask).cpu().item()
                rec = recall_(true_mask, pred_mask).cpu().item()
                dice = dice_score(true_mask, pred_mask).cpu().item()
                iou = jaccard_index(true_mask, pred_mask).cpu().item()

                # Record metrics
                metrics['accuracy'].append(acc)
                metrics['precision'].append(prec)
                metrics['recall'].append(rec)
                metrics['dice_score'].append(dice)
                metrics['IoU'].append(iou)

    return metrics
```

```
# Generate Segmentation Metrics
unet_metrics = compute_metrics(
    unet_model, test_loader, CFG.DEVICE
)
```

```
100%                                    24/24 [00:09<00:00,  2.58it/s]
```

```
# Create copy of test df
unet_test_df = test_df.copy()

# Concatenate Metrics onto copied df
unet_test_df = pd.concat(
    (unet_test_df, pd.DataFrame(unet_metrics)),
    axis=1
)

# Generate diagnosis labels
unet_test_df['model_diagnosis'] = [
    int(mask.max()) for mask in unet_test_masks
]
```

```
# View df
unet_test_df
```

	id	patient	image_path	mask_path	diagnosis	accuracy	precision	rec
0	22	TCGA_DU_A5TS_19970726	/kaggle/input/lgg-mri-segmentation/kaggle_3m/T...	/kaggle/input/lgg-mri-segmentation/kaggle_3m/T...	1	0.997250	0.747104	0.982
1	11	TCGA_DU_8165_19970205	/kaggle/input/lgg-mri-segmentation/kaggle_3m/T...	/kaggle/input/lgg-mri-segmentation/kaggle_3m/T...	0	1.000000	NaN	N
2	58	TCGA_HT_7881_19981015	/kaggle/input/lgg-mri-segmentation/kaggle_3m/T...	/kaggle/input/lgg-mri-segmentation/kaggle_3m/T...	0	1.000000	NaN	N
3	19	TCGA_DU_5853_19950823	/kaggle/input/lgg-mri-segmentation/kaggle_3m/T...	/kaggle/input/lgg-mri-segmentation/kaggle_3m/T...	0	1.000000	NaN	N
4	1	TCGA_DU_7298_19910324	/kaggle/input/lgg-mri-segmentation/kaggle_3m/T...	/kaggle/input/lgg-mri-segmentation/kaggle_3m/T...	0	1.000000	NaN	N
...	...	...	...	...	...	...	...	
762	29	TCGA_DU_6399_19830416	/kaggle/input/lgg-mri-segmentation/kaggle_3m/T...	/kaggle/input/lgg-mri-segmentation/kaggle_3m/T...	1	0.998266	0.864238	0.990
763	4	TCGA_DU_7294_19890104	/kaggle/input/lgg-mri-segmentation/kaggle_3m/T...	/kaggle/input/lgg-mri-segmentation/kaggle_3m/T...	0	1.000000	NaN	N
764	11	TCGA_HT_8105_19980826	/kaggle/input/lgg-mri-segmentation/kaggle_3m/T...	/kaggle/input/lgg-mri-segmentation/kaggle_3m/T...	0	1.000000	NaN	N
765	22	TCGA_DU_7014_19860618	/kaggle/input/lgg-mri-segmentation/kaggle_3m/T...	/kaggle/input/lgg-mri-segmentation/kaggle_3m/T...	1	0.997708	0.709677	0.747
766	58	TCGA_DU_6405_19851005	/kaggle/input/lgg-mri-segmentation/kaggle_3m/T...	/kaggle/input/lgg-mri-segmentation/kaggle_3m/T...	0	1.000000	NaN	N

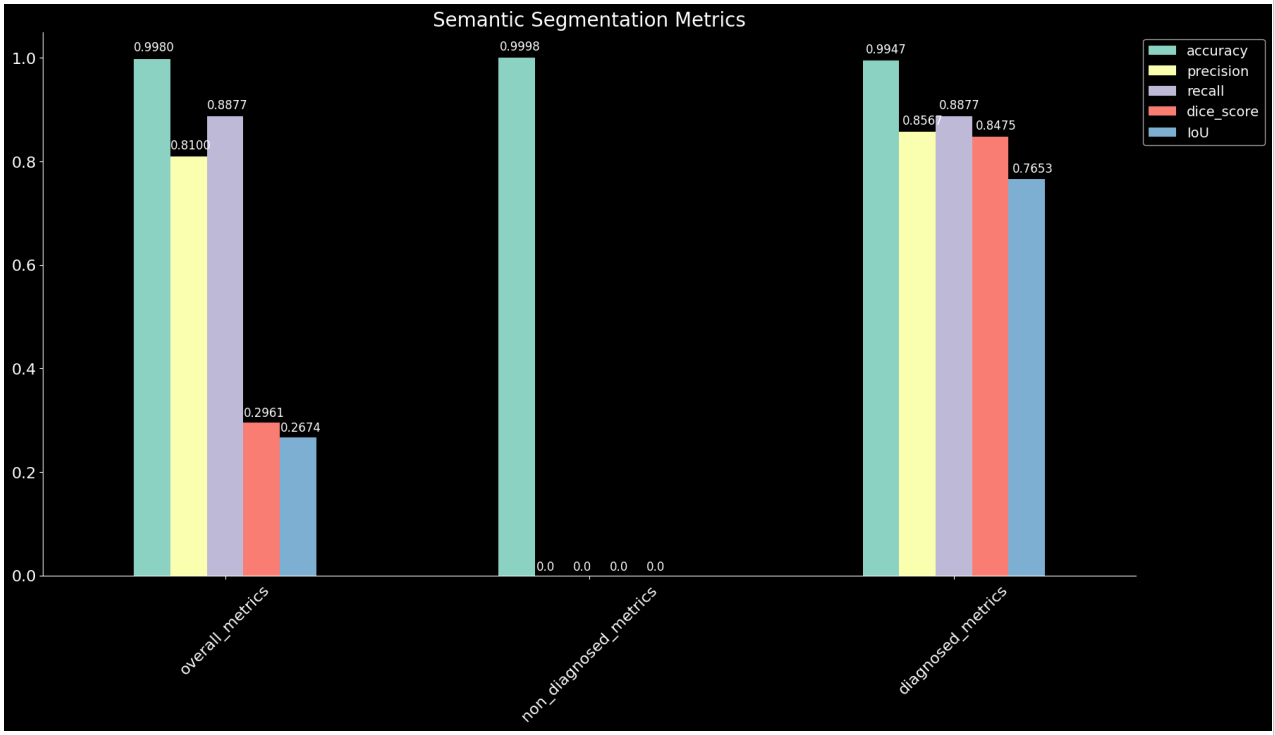
767 rows × 11 columns

```
# Get segmentation metrics for all samples
overall_metrics = dict(
    unet_test_df[unet_metrics.keys()]
    .mean()
)

# Get segmentation metrics for non-diagnosed samples
non_diagnosed_metrics = dict(
    unet_test_df[
        unet_test_df['diagnosis'] == 0
    ][unet_metrics.keys()].mean()
)

# Get segmentation metrics for diagnosed samples
diagnosed_metrics = dict(
    unet_test_df[
        unet_test_df['diagnosis'] == 1
    ][unet_metrics.keys()].mean()
)

# Compile metrics into dataframe
segmetation_metrics = pd.DataFrame({
    'overall_metrics': overall_metrics,
    'non_diagnosed_metrics': non_diagnosed_metrics,
    'diagnosed_metrics': diagnosed_metrics,
}).T

# View segmetation metrics df
segmetation_metrics
```

	accuracy	precision	recall	dice_score	IoU
overall_metrics	0.998035	0.810004	0.887684	0.296143	0.267419
non_diagnosed_metrics	0.999829	0.000000	NaN	0.000000	0.000000
diagnosed_metrics	0.994696	0.856735	0.887684	0.847543	0.765336

```
# Plot segmentation metrics bar plot
ax = segmetation_metrics.plot(
    kind="bar", rot=0,
    figsize=(20, 10)
)

# Add legend
ax.legend(bbox_to_anchor=(1.0, 1.0), fontsize=14)
```

```
# Display bar values
for p in ax.patches:
    bar_value = p.get_height()
    text = '0.0' if bar_value==0 else f'{bar_value:.4f}'
    ax.annotate(
        text,
        (p.get_x() * 1.005, p.get_height() * 1.005 + 0.01),
        fontsize=12
    )

# Update axes tick parameters
ax.tick_params(
    axis='both', which='major',
    labelsize=16
);

# Rotate x-axis ticks
ax.tick_params(
    axis='x', which='major',
    labelsize=16, rotation=45
);

# Add title & remove top and right borders w. sns.despine
plt.title('Semantic Segmentation Metrics', fontsize=20);
sns.despine();
```

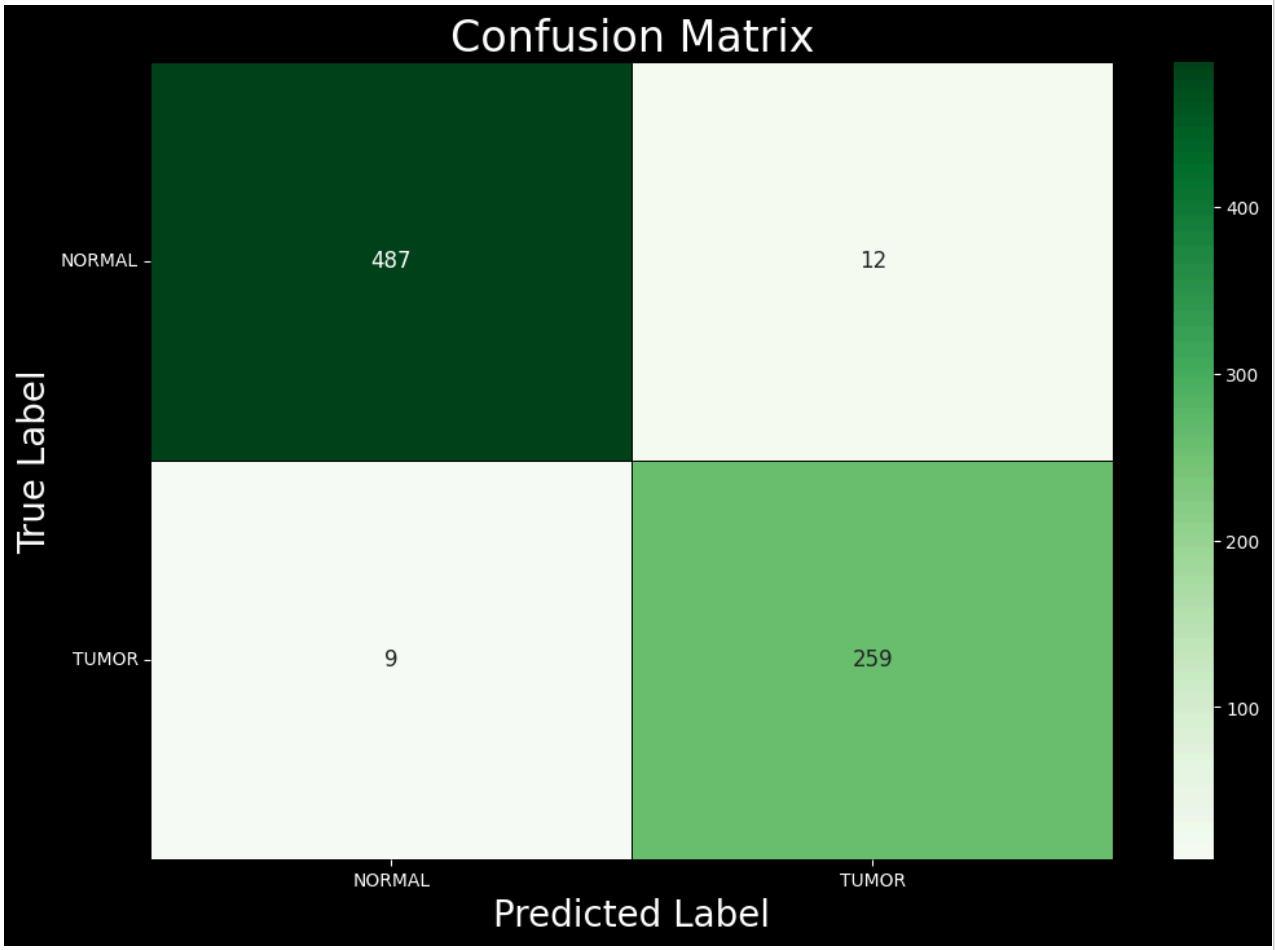



```
def plot_confusion_matrix(y_true, y_pred, classes='auto', figsize=(10, 10), text_size=12):
    # Generate confusion matrix
    cm = confusion_matrix(y_true, y_pred)

    # Set plot size
    plt.figure(figsize=figsize)

    # Create confusion matrix heatmap
    disp = sns.heatmap(
        cm, annot=True, cmap='Greens',
        annot_kws={"size": text_size}, fmt='g',
        linewidths=0.5, linecolor='black', clip_on=False,
        xticklabels=classes, yticklabels=classes)

    # Set title and axis labels
    disp.set_title('Confusion Matrix', fontsize=24)
```

```
    disp.set_xlabel('Predicted Label', fontsize=20)
    disp.set_ylabel('True Label', fontsize=20)
    plt.yticks(rotation=0)

    # Plot confusion matrix
    plt.show()

    return
```

```
# Get Test True Diagnosis Labels
test_labels = unet_test_df['diagnosis'].to_numpy()

# Get Test Predicted Diagnosis Labels
unet_predictions = unet_test_df['model_diagnosis'].to_numpy()
```

```
# Plot Confusion Matrix
plot_confusion_matrix(
    test_labels,
    unet_predictions,
    figsize=(12, 8),
    classes=['NORMAL', 'TUMOR']
)
```




```
print(
    classification_report(
        test_labels,
        unet_predictions,
        target_names=['NORMAL', 'TUMOR']
))
```

```
              precision    recall  f1-score   support

      NORMAL       0.98      0.98      0.98       499
       TUMOR       0.96      0.97      0.96       268

    accuracy                           0.97       767
   macro avg       0.97      0.97      0.97       767
weighted avg       0.97      0.97      0.97       767
```

```
def generate_performance_scores(y_true, y_pred, log=True):

    model_accuracy = accuracy_score(y_true, y_pred)
    model_precision, model_recall, model_f1, _ = (
        precision_recall_fscore_support(
            y_true, y_pred, average="weighted"
        )
    )
    model_matthews_corrcoef = matthews_corrcoef(y_true, y_pred)

    if log:
        print('=============================================')
        print(f'\nPerformance Metrics:\n')
        print('=============================================')
        print(f'accuracy_score:\t\t{model_accuracy:.4f}\n')
        print('_____________________________________________')
        print(f'precision_score:\t{model_precision:.4f}\n')
        print('_____________________________________________')
        print(f'recall_score:\t\t{model_recall:.4f}\n')
        print('_____________________________________________')
        print(f'f1_score:\t\t{model_f1:.4f}\n')
        print('_____________________________________________')
        print(f'matthews_corrcoef:\t{model_matthews_corrcoef:.4f}\n')
        print('=============================================')

    preformance_scores = {
        'accuracy_score'       : model_accuracy,
        'precision_score'      : model_precision,
        'recall_score'         : model_recall,
        'f1_score'             : model_f1,
        'matthews_corrcoef'    : model_matthews_corrcoef
    }
    return preformance_scores
```

```
# Generate U-Net model classification performance scores
unet_model_performance = generate_performance_scores(
    test_labels, unet_predictions, log=True
)
```

```
=============================================

Performance Metrics:

=============================================
accuracy_score:         0.9726

_____________________________________________
precision_score:        0.9727

_____________________________________________
recall_score:           0.9726

_____________________________________________
f1_score:               0.9727

_____________________________________________
matthews_corrcoef:      0.9400

=============================================
```

```
def view_multiple_predictions(
    model, ds, count=10,
    color_map='rgb',
    fig_size=(14, 10),
    title_size=16):

    # Select random samples
    idx = random.sample(
        list(range(len(ds))),
        count
    )

    # Initialise figure
    fig = plt.figure(figsize=fig_size);
    fig.tight_layout();
    fig.subplots_adjust(top=0.95);

    # Plot Samples
    for row, _ in enumerate(idx):
        image, mask = ds[_]
        idx = 3 * (row + 1)

        # Plot MRI Image
```

```
        plt.subplot(count+1, 3, idx + 1);
        plt.title(f'MRI Image', fontsize=title_size);
        if color_map=='rgb':
            plt.imshow(image.permute(1, 2, 0))
        else:
            plt.imshow(image.permute(1, 2, 0), cmap=color_map)

        # Plot Mask
        plt.subplot(count+1, 3, idx + 2);
        plt.title(f'True Mask', fontsize=title_size)

        gen_mask = np.dstack([mask[0]*0.1, mask[0]*0.45, mask[0]*0.1])
        plt.imshow(image.permute(1, 2, 0) + gen_mask);

        # Generate predicted mask
        mri_input = torch.unsqueeze(
            torch.tensor(image), dim=0
        ).to(CFG.DEVICE)

        # Set inferene context manager to active for predictions
        with torch.inference_mode():
            pred_mask = model(mri_input)
            pred_mask = torch.sigmoid(pred_mask).cpu()
            pred_mask = (pred_mask > 0.5) #threshold of 0.5

        # Plot Predicted Mask
        plt.subplot(count+1, 3, idx + 3);
        plt.title(f'Predicted Mask', fontsize=title_size)

        pred_gen_mask = np.dstack([pred_mask[0][0]*0.3, pred_mask[0][0]*0.2, pred_mask[0][0]*0.8])
        plt.imshow(image.permute(1, 2, 0)+ pred_gen_mask);

    return
```

```
# Create Dataset for Diagnosed Sample
test_diagnosed_ds = MRISegmetationDataset(
    unet_test_df[
        unet_test_df['diagnosis'] == 1
    ].reset_index(drop=True),
    transform=inference_transforms
)
```

```
# View Random Samples
view_multiple_predictions(
    unet_model, test_diagnosed_ds,
    count=8, color_map='rgb',
    fig_size=(20, 38)
)
```

```
WARNING:matplotlib.image:Clipping input data to the valid range for imshow with RGB data ([0..1] for floats or
```